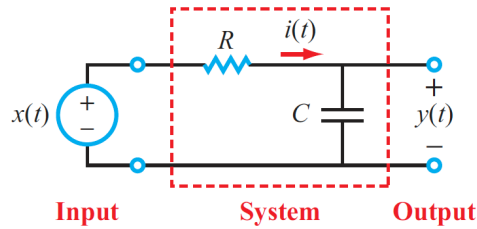


ECE 3337: Signals & Systems Analysis, Class Worksheet 5

1. (30 points) Consider the following resistor - capacitor (RC) circuit,



(a) RC circuit

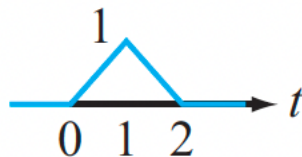
- (15 points) Use the convolution integral to re-derive a formula for the ramp response $y_{RAMP}(t)$ for this circuit using the formula for the impulse response of this circuit $h(t) = \frac{1}{RC} e^{-t/RC} u(t)$. Make sure to show your derivation steps.
- (15 points) Derive a formula for the response of the above circuit for the following input signal:

$$x(t) = r(t) - r(t - 1) - r(t - 2) + r(t - 3)$$

2. (40 points) Calculate the following convolution integrals. Make sure to show all of the detailed steps, especially calculation of the integration limits and the range of validity of time.

- $y_1(t) = u(t - 2) * u(t - 1)$
- $y_2(t) = \delta(t - 1) * r(t - 2)$
- $y_3(t) = e^{-at} u(t) * u(t - 4)$
- $y_4(t) = e^{-2t} u(t) * e^{-2t} u(t)$

3. (30 points) The step response of a system is plotted below:



- (15 points) Calculate the impulse response of this system
- (15 points) Calculate the response of this same system to the following input. Make sure to show all of the detailed steps.

